

WEEK 8: ORGAN SYSTEM

RESPIRATORY SYSTEM

FUNCTIONS OF THE RESPIRATORY SYSTEM

- Respiration includes the following processes:
 - Exchange of O₂ and CO₂ between the air in the lungs and the blood
 - Transport of O₂ and CO₂ in the blood and tissues
 - Regulation of blood pH by changing blood CO₂ levels
 - Voice production by air movement passing the vocal cords
 - Olfaction occurs when the airborne molecules are drawn into the nasal cavity
 - Innate immunity by prevent the pathogens from entering the body and by removing them from respiratory surfaces

ANATOMY OF THE RESPIRATORY SYSTEM

- Divided into upper and lower respiratory tract
 - **Upper:** external nose, nasal cavity, pharynx (throat), paranasal sinuses (**NEPP**)
 - **Lower:** larynx, trachea, bronchi, bronchioles, alveoli, lungs, (**BBALT**)
- Functional division:
 - **Conducting zone** (nose -> terminal bronchioles) - passageway for air, humification, warming, and filtration
 - **Respiratory zone** - (respiratory bronchioles -> alveolar ducts) - gas exchange

NOSE

- Warms, filters, and moistens the air; resonance for speech; olfaction

- Consists of external nose and nasal cavity:

PENNNCCH

- **External nose** is the visible structure that forms a prominent feature of the face
 - Most are consists of hyaline cartilage, although the bridge consists of bones
- **Nares (nostrils)** are external opening of the nose

- **Choanae** are the opening into the pharynx
- **Nasal cavity** extends from the nares to the choanae
 - Filled with respiratory mucosa (pseudostratified ciliated columnar epithelium for sneezing reflex)
- **Nasal septum** is a partition dividing the nasal cavity into left and right parts
 - Deviated nasal septum occurs when the septum bulges to one side
- **Hard palate** forms the floor of the nasal cavity, separating the nasal cavity from oral cavity
- **Conchae** are three prominent bone ridges present on the lateral walls on each side of the nasal cavity
 - Increases surface area of the nasal cavity and cause air to churn, so that it can be cleansed, humidified, and warmed
- **Paranasal sinuses** are air-filled spaces within bone
 - Includes maxillary, frontal, ethmoidal, and sphenoidal sinuses
 - Lighten the skull, produce mucus, and influence the quality of the voice by acting as resonating chambers
 - Meatus - draining chamber for sinuses
 - Passageway for infection
- Air passage through the nasal cavity
 - Air enters via nares (nostrils), which inside is lined with stratified squamous epithelium that contains coarse hairs to trap large dust particles
 - The rest of nasal cavity is lines with pseudostratified ciliated columnar epithelium, which contains cilia for sweeping debris-laden mucus posteriorly, and goblet cells for producing mucus that traps smaller particles
 - The mucus with trapped apticles is transported to the phrarynx by ciliary action. Once in the pharynx, mucus is swallowed,

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sending debris to the digestive tract for removal

- As air passes through nasal cavities, it will be humidified by moisture from mucous epithelium, and warmed by heat exchange from superficial capillary networks

- **Sneeze reflex** - dislodges foreign substances from the nasal cavity

PHARYNX

- Most common passageway for both air and food

- Muscular tube

- Air from the nasal cavity and air, food, and water from the mouth passes through the pharynx

- Divided into **nasopharynx**, **oropharynx**, and **laryngopharynx**

- **Nasopharynx** - *superior part of the pharynx*
 - Located **posterior to the choanae** and **superior to the soft palate**, which is an incomplete muscle and connective tissue partition separating nasopharynx from oropharynx
 - Soft palate is elevated during swallowing; closes nasopharynx; contains pharyngeal tonsil and eustachian tube opening (clogged nose - affected ears)
 - Lined with pseudostratified ciliated columnar epithelium, continuous with the nasal cavity
 - **Uvula** is the posterior extension of the soft palate
 - **Auditory tubes** extend from the middle ears and open into the nasopharynx
 - **Pharyngeal tonsil** forms the posterior part, which helps defend the body against infection
- **Oropharynx**

- Extends from the uvula to the epiglottis, and posterior to the oral cavity
 - Thus, food, drink, and air pass through here
- Lined with stratified squamous epithelium, which protects against abrasion
- **Palatine tonsils** are located near the opening between the mouth and the oropharynx
 - Located in the lateral walls near the border of the oral cavity and the oropharynx

- **Laryngopharynx**

- Connects to esophagus and larynx
- Lined with stratified squamous epithelium and ciliated columnar epithelium
- Food and drink pass through, with a small amount of air
 - Swallowing too much air can cause excess gas in the stomach and may result in belching

LARYNX

- Located in the anterior throat and extends from the base of the tongue to the trachea

- Passageway for air between pharynx and trachea, necessary for sound production, patency of pathway

- **Consists of an outer casing of 9 cartilages** connected to one another by muscles and ligaments

- 3 are unpaired, 6 forms 3 pairs

- **Largest** - unpaired thyroid cartilage (Adam's apple)
 - Attached superiorly to the hyoid bone
 - Two pairs of ligaments extend from the posterior surface: **vestibular folds** (superior: false vocal cords)

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and **vocal folds** (inferior: true vocal cords)

- **Vestibular folds** prevent food and liquids from entering pharynx, like epiglottis
- **Vocal folds** - voice production
 - **Laryngitis** - inflammation of the mucous epithelium of the vocal folds

- **Most inferior** - unpaired cricoid cartilage
 - Forms the base of the larynx on which the other cartilages rest
- **Third unpaired** - epiglottis
 - Differs because it consists of elastic rather than hyaline cartilage
 - Helps prevent swallowed materials from entering the larynx
 - As the larynx elevates during swallowing, the epiglottis tips posteriorly to cover the opening of the larynx

- **6 paired cartilages consist of 3 cartilages** on each side of the posterior part of the larynx, which forms an attachment site for the vocal cords

- **Cuneiform** - top cartilage of each side
- **Corniculate** - middle
- **Arytenoid** - articulate with the cricoid cartilage inferiorly

TRACHEA - "windpipe"

- 10-12 cm tube at T5 level

- Membranous tube attached to the larynx, which consists of connective tissue and smooth muscles, reinforced with **16-20 C-shaped pieces of hyaline cartilage**

- Posterior wall has no cartilage and consists of a ligamentous membrane and smooth muscle
- Smooth muscle can alter the diameter of the trachea

- Sensory receptors detect the foreign substances, and APs travel along the vagus nerves to the medulla oblongata. where the reflex is triggered
- **Posterior walls has trachealis muscle for cough reflex**

- Begins immediately inferior to the cricoid cartilage
- It projects through the mediastinum and divides into the right and left primary bronchi at the level of the T5

- Esophagus lies immediately posterior to the trachea

- Lined with pseudostratified ciliated columnar epithelium, which contains cilia and goblet cells

- Cilia propel mucus produced by the goblet cells through larynx and into the pharynx
- Irritation from cigarette smoking causes it to turn to stratified squamous epithelium (no cilia so it cannot clear the airway)
 - **Accumulations of mucus = cough reflex = smoker's cough**

BRONCHI

- Trachea divides into the left and right main or primary bronchi, each of which connects to a lung

- **Left main bronchus is more horizontal** than the right main bronchus because it is displaced by the heart
- **Right main bronchus is more vertical**, wider, and shorter; common site where aspirated objects are directed

- Line with pseudostratified ciliated columnar epithelium and are supported by C-shaped pieces of cartilage

LUNGS

- Principle organ for respiration

- **Outer surface:** pleura

- **Visceral:** lines the lung
- **Parietal:** lines the thoracic wall
- **Pleural cavity:** maintains negative pressure and also lubricates

- Located in the thoracic cavity

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- Each lung is cone-shaped, with its base resting on the diaphragm and its apex extending superiorly to a point about 2.5 cm above the clavicle

- **Right lung has 3 lobes:** superior, middle, and inferior lobes (10 BC segments)

- **Horizontal fissure** - separates superior and middle
- **Oblique fissure** - separates middle and inferior

- **Left lung has 2 lobes:** superior and inferior lobes (8-10 BC segments)

- Contains lingula

- Main bronchi branch many times form the tracheobronchial tree

- Each main bronchus divides into lobar bronchi as they enter their respective lungs
- The lobar bronchi (secondary bronchi), 2 in the left lung and 3 in the right lung, conduct air to each lobe
- The lobar bronchi in turn give rise to segmental bronchi (tertiary bronchi), which extend to the BC segments of the lungs
- The bronchi continue to branch many times, finally giving rise to bronchioles - no cartilage, smooth muscle regulation

- Bronchioles also subdivide numerous times to give rise to terminal bronchioles, to respiratory bronchioles

- Each respiratory bronchiole subdivides to form alveolar ducts, which are life long, branching hallways with many open doorways
- The doorways open into alveoli, which are small air sacs
 - Functional unit
 - ~300 million in adults; provide 70 m² surface area
 - Cell type:
 - Type I - gas exchange
 - Type II - produce surfactant (reduces surface tension, friction, and prevent collapse)
 - Alveolar macrophage (dust cell) - immune defense

- Alveolar ducts end as 2 or 3 alveolar sacs, which are chambers connected to 2 or more alveoli

- **Respiratory membrane** - where gas exchange between air and blood takes place

- Formed mainly by the walls of the alveoli and surrounding capillaries, but alveolar ducts and respiratory bronchioles also contribute
- Alveolar epithelium (Type I pneumonocytes), capillary endothelium, and fused basement membrane

- **Surfactant** - chemical secreted by the specialized secretory cells within the walls of the alveoli, which reduces the tendency of alveoli to recoil

- Relaxation and contraction of the smooth muscle within the bronchi and bronchioles can change the diameter of the air passageways

- E.g., during exercise the diameter can increase, thus increasing the volume of air moved
- E.g., during asthma attack, however, contraction of the smooth muscle in the terminal bronchioles can result in greatly reduced airflow
 - **Albuterol** help counteract an asthma attack by promoting smooth muscle relaxation in the wall of terminal bronchioles

PLEURAL CAVITIES

- Contained within the thoracic cavity

- **Pleura** - serous membrane that lines the pleural cavity (the lungs itself)

- Consists of parietal and visceral pleura
- **Parietal** - lines the walls of the thorax, diaphragm, and mediastinum
- **Visceral** - covers the surface of the lung
 - **Pleural fluid** - small volume of pleural fluid found between the two, which is produced by the pleural membrane
 - It acts a lubricant for visceral and parietal pleurae to slide

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past each other as the lungs and thorax change shape during respiration

- it helps hold the pleural membranes together

LYMPHATIC SUPPLY

- Lymphatic drainage - drains excess lymph

- Two lymphatic supply of the lungs:

- **Superficial lymphatic vessels**
 - Drain lymph from the **superficial lung tissue** (surface) and the **visceral pleural**
- **Deep lymphatic vessels**
 - Drain lymph from the **bronchi vessels** and **associated connective tissues**
 - **Both lymphatic vessels exit the lungs at the main bronchi**

- No lymphatic vessels are located in the walls of the alveoli

- **Clinical relevance:** lung cancer metastasis

Sequence of lymphatic drainage:

Lymph from superficial and deep lymphatic plexus

-> hilar (bronchopulmonary) -> tracheobronchial nodes -> paratracheal -> bronchomediastinal trunks (nodes)

MUSCLES OF PULMONARY VENTILATION

- **Muscles of inspiration:**

- **Diaphragm** - large dome of the skeletal muscle that separates the thoracic cavity from the abdominal cavity
 - Nerve: phrenic (C3-C5)
- **External intercostalis** - muscles that elevate the ribs and sternum (also thoracic vertebrae); the muscles are comprised of:
 - **SUPAS** - SCM, upper trapezius, pectoralis minor, serratus anterior posterior, scalene

- **Forced inspiration:** SCM and upper trapezius

- **Muscle of expiration:**

- **NO MUSCLE** = passive (elastic recoil)
- If forced expiration:
 - **Internal intercostalis and abdominal muscle** - depress the ribs and sternum (also thoracic vertebrae)

- At the end of a normal, quiet expiration, the respiratory muscles are relaxed

- During quiet inspiration, contraction of the diaphragm cause the top of the dome to move inferiorly, which increases the volume of the thoracic cavity

PRESSURE CHANGES AND AIRFLOW

- As the volume of a container increases, the pressure within the container also increases, and vice versa

- Air flows from an area of higher pressure to an area of lower pressure

- The greater the pressure difference the greater the rate of airflow

1.) At the end of expiration

- Pressure within the alveoli, is equal to atm pressure, which is the air pressure outside the body

- No air moves in or out of the lungs

2.) During inspiration

- Contraction of the muscles of inspiration increases the volume of thoracic cavity

- Increased thoracic volume causes the lungs to expand, resulting in alveolar volume increase
- Alveolar volume increase while atmospheric pressure decrease

3.) At the end of inspiration

- Thorax and alveoli stop expanding

- Alveolar pressure and atm pressure become equal (airflow stops)

4.) During expiration

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- Thoracic volume decreases, which also decreases alveolar volume
 - Alveolar pressure increases above atm pressure

2.) *Parasympathetic fibers* (vagus nerve) - bronchoconstriction, mucus secretion

3.) *Pulmonary plexus* - automatic innervation for airway tone and reflexes

BLOOD FLOW

1.) *Pulmonary circulation*

- RV to PA -> deoxygenated
- PV to LA -> oxygenated

2.) *Bronchial circulation*

- Bronchial arteries (from thoracic aorta) -> supply blood to lung tissue (except alveoli)
 - Except alveoli because it supply its own

VENOUS DRAINAGE

1.) *Pulmonary venous drainage*

- PV brings oxygenated blood to LA
- Each lung drains to two different PV
 - Right lung: left superior (superior and middle lobes) and inferior (inferior lobe) PV
 - Left lung: right superior (superior lobe and lingula) and inferior (inferior lobe) PV
- PV collect blood from the capillary beds of alveoli through venules and progressively larger tributaries, and finally to PV

2.) *Bronchial venous drainage*

- Supply oxygen and nutrients to the lung tissue
- Venous drainage:
 - Bronchial veins: drains blood from the walls of the bronchi and connective tissue
- Drainage pattern:
 - Left bronchial vein - drains into accessory hemiazygos vein of the left superior intercostal vein
 - Right bronchial vein - drains into azygos veins

NERVOUS SYSTEM SUPPLY

- 1.) *Sympathetic fibers*** - bronchodilation